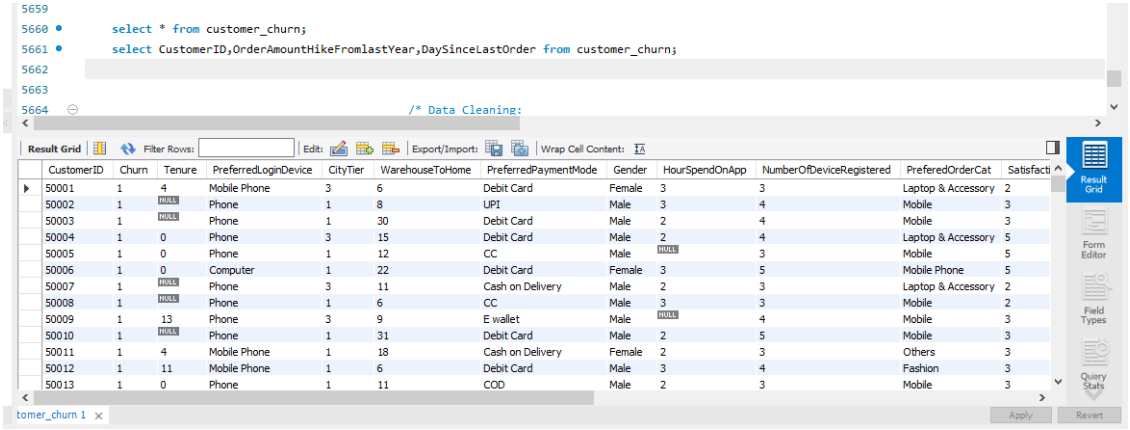
E-Commerce Customer Churn Analysis

Database Name: ecomm

Table Name: customer_churn



```
5659
5660 •    select * from customer_churn;
5661 •    select CustomerID,OrderAmountHikeFromlastYear,DaySinceLastOrder from customer_churn;
5662
5663
5664                                        /* Data Cleaning:
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

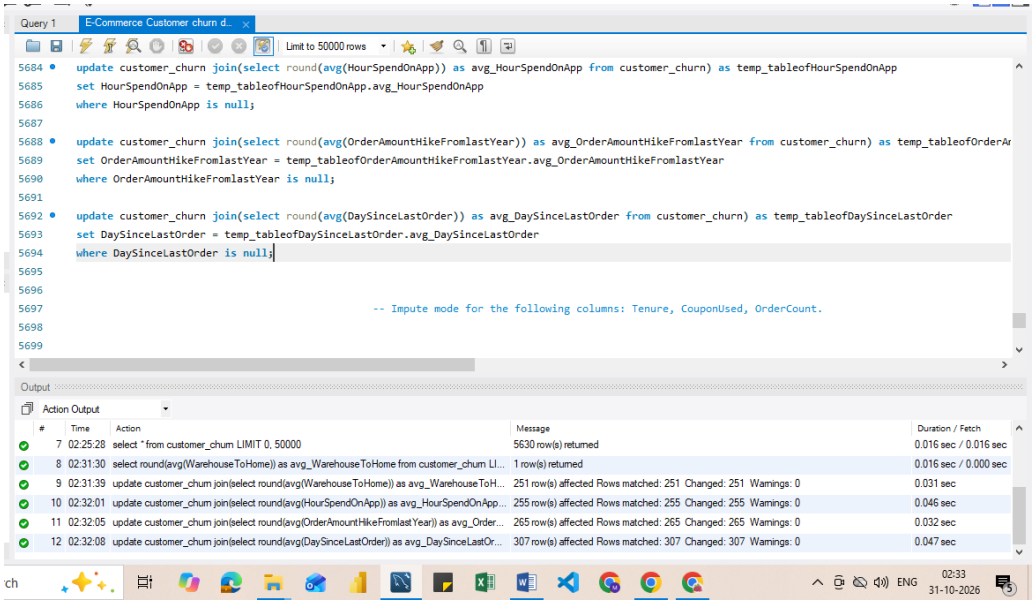
CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrderCat	Satisfacti
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Phone	1	12	CC	Male	NULL	3	Mobile	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1	NULL	Phone	1	6	CC	Male	3	3	Mobile	2
50009	1	13	Phone	3	9	E wallet	Male	NULL	4	Mobile	3
50010	1	NULL	Phone	1	31	Debit Card	Male	2	5	Mobile	3
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion	3
50013	1	0	Phone	1	11	COD	Male	2	3	Mobile	3

tomer_churn 1 | Apply | Revert

Data Cleaning:

Handling Missing Values and Outliers:

➢ Impute mean for the following columns, and round off to the nearest integer if required: WarehouseToHome, HourSpendOnApp, OrderAmountHikeFromlastYear, DaySinceLastOrder.



Query 1 | E-Commerce Customer churn d... | Limit to 50000 rows

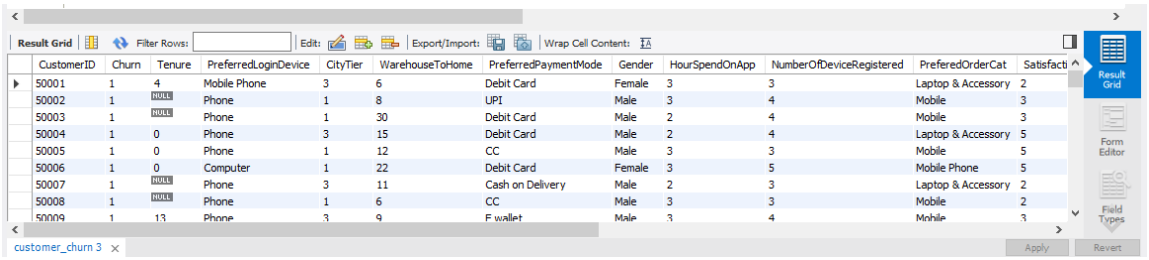
```
5684 •  update customer_churn join(select round(avg(HourSpendOnApp)) as avg_HourSpendOnApp from customer_churn) as temp_tableofHourSpendOnApp
5685    set HourSpendOnApp = temp_tableofHourSpendOnApp.avg_HourSpendOnApp
5686    where HourSpendOnApp is null;
5687
5688 •  update customer_churn join(select round(avg(OrderAmountHikeFromlastYear)) as avg_OrderAmountHikeFromlastYear from customer_churn) as temp_tableofOrderAr
5689    set OrderAmountHikeFromlastYear = temp_tableofOrderAmountHikeFromlastYear.avg_OrderAmountHikeFromlastYear
5690    where OrderAmountHikeFromlastYear is null;
5691
5692 •  update customer_churn join(select round(avg(DaySinceLastOrder)) as avg_DaySinceLastOrder from customer_churn) as temp_tableofDaySinceLastOrder
5693    set DaySinceLastOrder = temp_tableofDaySinceLastOrder.avg_DaySinceLastOrder
5694    where DaySinceLastOrder is null;
5695
5696
5697                                        -- Impute mode for the following columns: Tenure, CouponUsed, OrderCount.
5698
5699
```

Output | Action Output

#	Time	Action	Message	Duration / Fetch
7	02:25:28	select * from customer_churn LIMIT 0, 50000	5630 row(s) returned	0.016 sec / 0.016 sec
8	02:31:30	select round(avg(WarehouseToHome)) as avg_WarehouseToHome from customer_churn LI...	1 row(s) returned	0.016 sec / 0.000 sec
9	02:31:39	update customer_churn join(select round(avg(WarehouseToHome)) as avg_WarehouseToH...	251 row(s) affected Rows matched: 251 Changed: 251 Warnings: 0	0.031 sec
10	02:32:01	update customer_churn join(select round(avg(HourSpendOnApp)) as avg_HourSpendOnApp...	255 row(s) affected Rows matched: 255 Changed: 255 Warnings: 0	0.046 sec
11	02:32:05	update customer_churn join(select round(avg(OrderAmountHikeFromlastYear)) as avg_Order...	265 row(s) affected Rows matched: 265 Changed: 265 Warnings: 0	0.032 sec
12	02:32:08	update customer_churn join(select round(avg(DaySinceLastOrder)) as avg_DaySinceLastOr...	307 row(s) affected Rows matched: 307 Changed: 307 Warnings: 0	0.047 sec

ENG 02:33 31-10-2026

Checking the table:



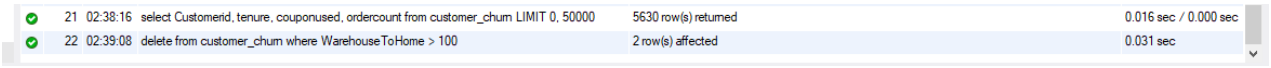
CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrderCat	Satisfacti
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	NULL	Phone	1	8	UPI	Male	3	4	Mobile	3
50003	1	NULL	Phone	1	30	Debit Card	Male	2	4	Mobile	3
50004	1	0	Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Phone	1	12	CC	Male	3	3	Mobile	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1	NULL	Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1	NULL	Phone	1	6	CC	Male	3	3	Mobile	2
50009	1	13	Phone	3	9	E wallet	Male	3	4	Mobile	3

➢ Impute mode for the following columns: Tenure, CouponUsed, OrderCount.



Customerid	tenure	couponused	ordercount
50001	4	1	1
50002	1	0	1
50003	1	0	1
50004	0	0	1
50005	0	1	1
50006	0	4	6
50007	1	0	1
50008	1	2	2
50009	13	0	1
50010	1	1	1

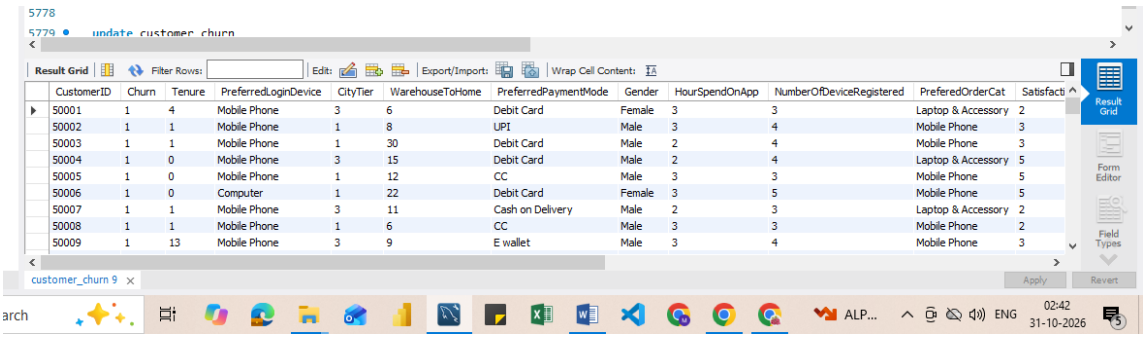
➢ Handle outliers in the 'WarehouseToHome' column by deleting rows where the values are greater than 100.



21	02:38:16	select Customerid, tenure, couponused, ordercount from customer_churn LIMIT 0, 50000	5630 row(s) returned	0.016 sec / 0.000 sec
22	02:39:08	delete from customer_churn where WarehouseToHome > 100	2 row(s) affected	0.031 sec

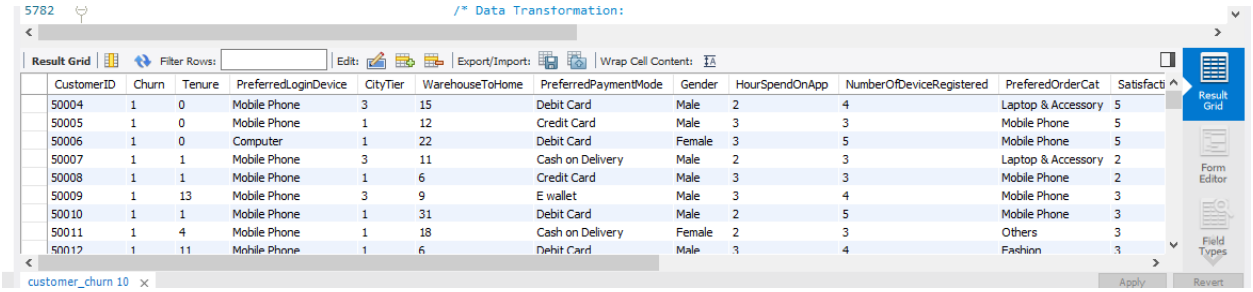
Dealing with Inconsistencies:

➢ Replace occurrences of “Phone” in the 'PreferredLoginDevice' column and “Mobile” in the 'PreferedOrderCat' column with “Mobile Phone” to ensure uniformity.



CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrderCat	Satisfacti
50001	1	4	Mobile Phone	3	6	Debit Card	Female	3	3	Laptop & Accessory	2
50002	1	1	Mobile Phone	1	8	UPI	Male	3	4	Mobile Phone	3
50003	1	1	Mobile Phone	1	30	Debit Card	Male	2	4	Mobile Phone	3
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Mobile Phone	1	12	CC	Male	3	3	Mobile Phone	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1	1	Mobile Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1	1	Mobile Phone	1	6	CC	Male	3	3	Mobile Phone	2
50009	1	13	Mobile Phone	3	9	E wallet	Male	3	4	Mobile Phone	3

➢ Standardize payment mode values: Replace "COD" with "Cash on Delivery" and "CC" with "Credit Card" in the PreferredPaymentMode column.



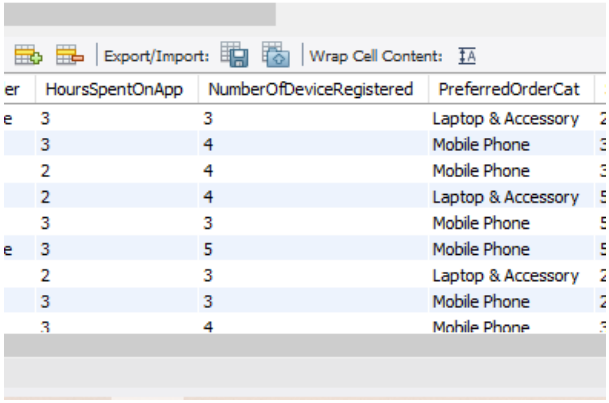
CustomerID	Churn	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HourSpendOnApp	NumberOfDeviceRegistered	PreferedOrderCat	Satisfacti
50004	1	0	Mobile Phone	3	15	Debit Card	Male	2	4	Laptop & Accessory	5
50005	1	0	Mobile Phone	1	12	Credit Card	Male	3	3	Mobile Phone	5
50006	1	0	Computer	1	22	Debit Card	Female	3	5	Mobile Phone	5
50007	1	1	Mobile Phone	3	11	Cash on Delivery	Male	2	3	Laptop & Accessory	2
50008	1	1	Mobile Phone	1	6	Credit Card	Male	3	3	Mobile Phone	2
50009	1	13	Mobile Phone	3	9	E wallet	Male	3	4	Mobile Phone	3
50010	1	1	Mobile Phone	1	31	Debit Card	Male	2	5	Mobile Phone	3
50011	1	4	Mobile Phone	1	18	Cash on Delivery	Female	2	3	Others	3
50012	1	11	Mobile Phone	1	6	Debit Card	Male	3	4	Fashion	3

Data Transformation:

Column Renaming:

➢ Rename the column "PreferedOrderCat" to "PreferredOrderCat".

➢ Rename the column "HourSpendOnApp" to "HoursSpentOnApp".

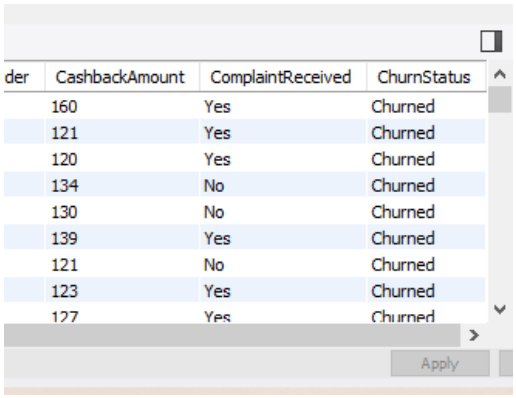


er	HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat	
e	3	3	Laptop & Accessory	2
	3	4	Mobile Phone	3
	2	4	Mobile Phone	3
	2	4	Laptop & Accessory	5
	3	3	Mobile Phone	5
e	3	5	Mobile Phone	5
	2	3	Laptop & Accessory	2
	3	3	Mobile Phone	2
	3	4	Mobile Phone	3

Creating New Columns:

➢ Create a new column named ‘ComplaintReceived’ with values "Yes" if the corresponding value in the ‘Complain’ is 1, and "No" otherwise.

➢ Create a new column named 'ChurnStatus'. Set its value to “Churned” if the corresponding value in the 'Churn' column is 1, else assign “Active”.

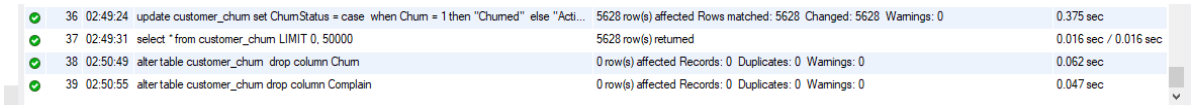


der	CashbackAmount	ComplaintReceived	ChurnStatus
	160	Yes	Churned
	121	Yes	Churned
	120	Yes	Churned
	134	No	Churned
	130	No	Churned
	139	Yes	Churned
	121	No	Churned
	123	Yes	Churned
	127	Yes	Churned

Apply

Column Dropping:

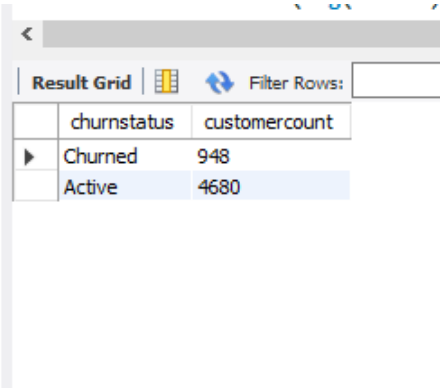
➢ Drop the columns "Churn" and "Complain" from the table.



36	02:49:24	update customer_churn set ChurnStatus = case when Churn = 1 then "Churned" else "Acti...	5628 row(s) affected Rows matched: 5628 Changed: 5628 Warnings: 0	0.375 sec
37	02:49:31	select * from customer_churn LIMIT 0, 50000	5628 row(s) returned	0.016 sec / 0.016 sec
38	02:50:49	alter table customer_churn drop column Churn	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.062 sec
39	02:50:55	alter table customer_churn drop column Complain	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec

Data Exploration and Analysis:

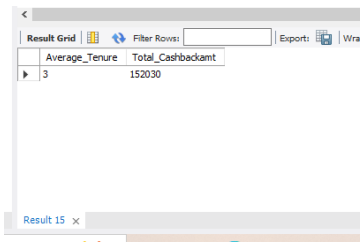
➢ Retrieve the count of churned and active customers from the dataset.



Result Grid | Filter Rows:

churnstatus	customercount
Churned	948
Active	4680

➢ Display the average tenure and total cashback amount of customers who churned.



Result Grid | Filter Rows: | Export: | Wra

Average_Tenure	Total_Cashbackamt
3	152030

Result 15

➢ Determine the percentage of churned customers who complained.

```
5859    from customer_churn
5860    where churnstatus = "churn
5861
```

Result Grid | Filter Rows:

complaint_percentage
53.59

Result 18

➢ Identify the city tier with the highest number of churned customers whose preferred order category is Laptop & Accessory.

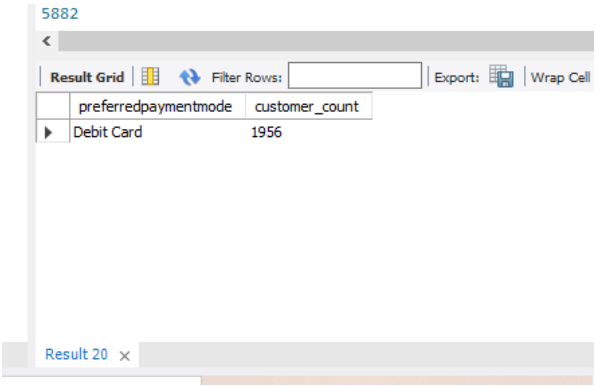
```
5876    select preferredpaymentmode, count(*) as cust
```

Result Grid | Filter Rows: | Export: | Wr

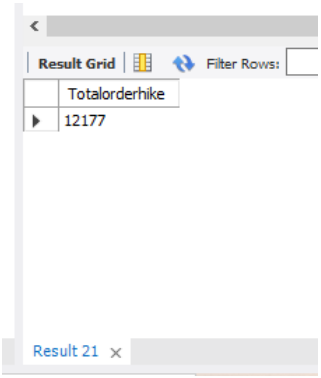
citytier	preferredordercat	churnedcustomers
3	Laptop & Accessory	150

Result 19

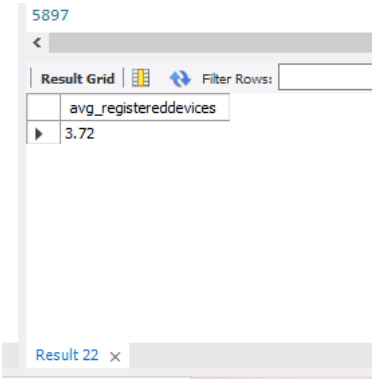
➢ Identify the most preferred payment mode among active customers.



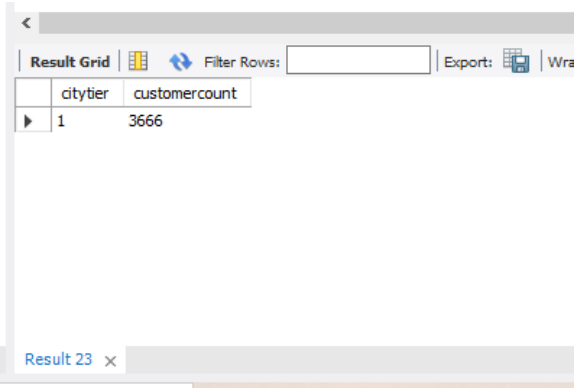
➢ Calculate the total order amount hike from last year for customers who are single and prefer mobile phones for ordering.



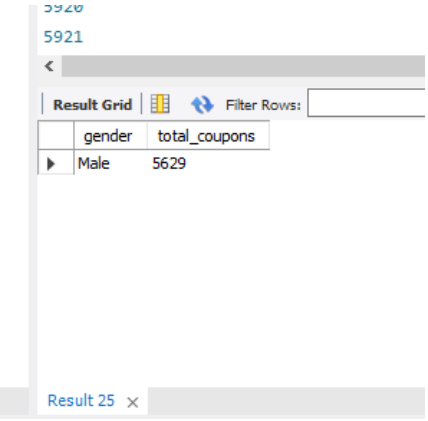
➢ Find the average number of devices registered among customers who used UPI as their preferred payment mode.



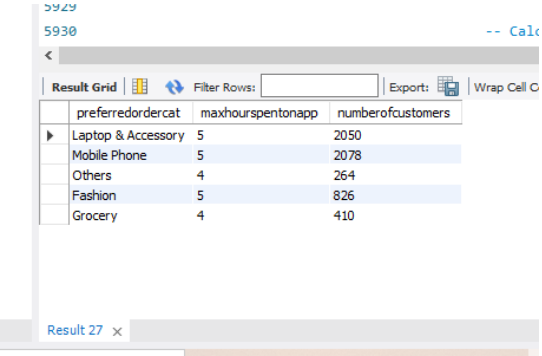
➢ Determine the city tier with the highest number of customers.



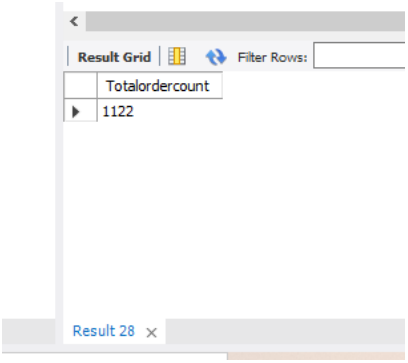
➢ Identify the gender that utilized the highest number of coupons.



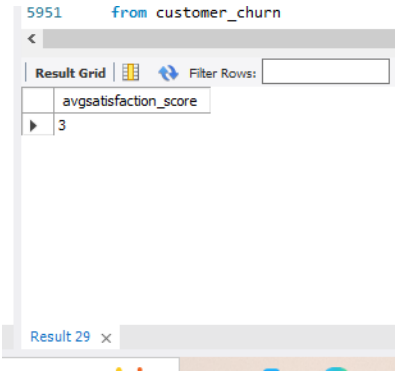
➢ List the number of customers and the maximum hours spent on the app in each preferred order category.



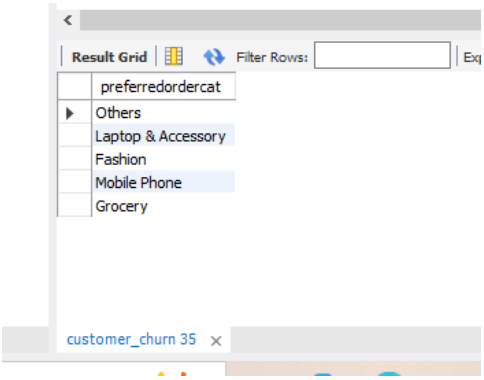
➢ Calculate the total order count for customers who prefer using credit cards and have the maximum satisfaction score.



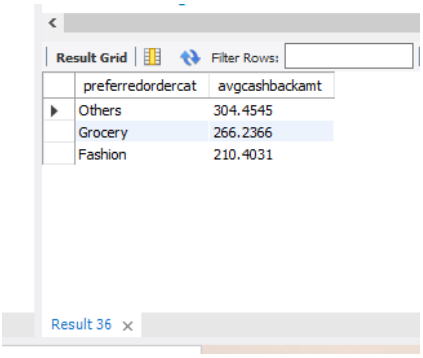
➢ What is the average satisfaction score of customers who have complained?



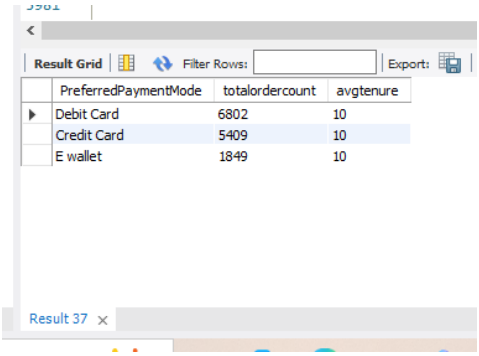
➢ List the preferred order category among customers who used more than 5 coupons.



➢ List the top 3 preferred order categories with the highest average cashback amount.



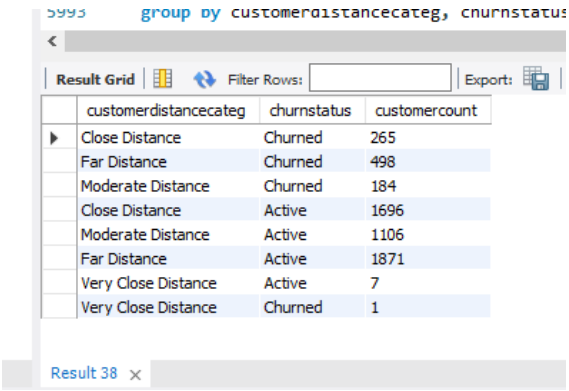
➢ Find the preferred payment modes of customers whose average tenure is 10 months and have placed more than 500 orders.



PreferredPaymentMode	totalordercount	avgtenure
Debit Card	6802	10
Credit Card	5409	10
E wallet	1849	10

Result 37

➢ Categorize customers based on their distance from the warehouse to home such as 'Very Close Distance' for distances <=5km, 'Close Distance' for <=10km, 'Moderate Distance' for <=15km, and 'Far Distance' for >15km. Then, display the churn status breakdown for each distance category.

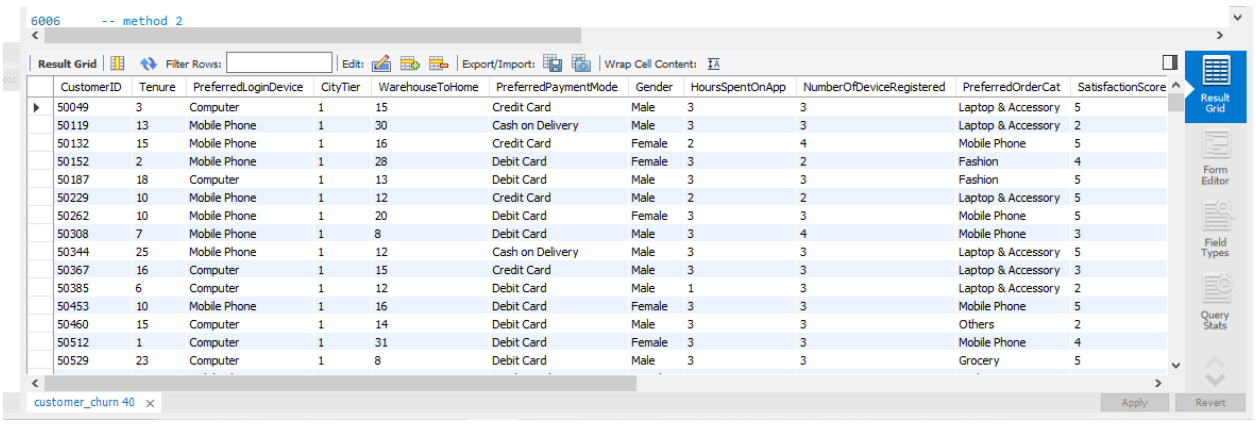


```
group by customerdistancecateg, churnstatus
```

customerdistancecateg	churnstatus	customercount
Close Distance	Churned	265
Far Distance	Churned	498
Moderate Distance	Churned	184
Close Distance	Active	1696
Moderate Distance	Active	1106
Far Distance	Active	1871
Very Close Distance	Active	7
Very Close Distance	Churned	1

Result 38

➢ List the customer's order details who are married, live in City Tier-1, and their order counts are more than the average number of orders placed by all customers.



```
-- method 2
```

CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDeviceRegistered	PreferredOrderCat	SatisfactionScore
50049	3	Computer	1	15	Credit Card	Male	3	3	Laptop & Accessory	5
50119	13	Mobile Phone	1	30	Cash on Delivery	Male	3	3	Laptop & Accessory	2
50132	15	Mobile Phone	1	16	Credit Card	Female	2	4	Mobile Phone	5
50152	2	Mobile Phone	1	28	Debit Card	Female	3	2	Fashion	4
50187	18	Computer	1	13	Debit Card	Male	3	3	Fashion	5
50229	10	Mobile Phone	1	12	Credit Card	Male	2	2	Laptop & Accessory	5
50262	10	Mobile Phone	1	20	Debit Card	Female	3	3	Mobile Phone	5
50308	7	Mobile Phone	1	8	Debit Card	Male	3	4	Mobile Phone	3
50344	25	Mobile Phone	1	12	Cash on Delivery	Male	3	3	Laptop & Accessory	5
50367	16	Computer	1	15	Credit Card	Male	3	3	Laptop & Accessory	3
50385	6	Computer	1	12	Debit Card	Male	1	3	Laptop & Accessory	2
50453	10	Mobile Phone	1	16	Debit Card	Female	3	3	Mobile Phone	5
50460	15	Computer	1	14	Debit Card	Male	3	3	Others	2
50512	1	Computer	1	31	Debit Card	Female	3	3	Mobile Phone	4
50529	23	Computer	1	8	Debit Card	Male	3	3	Grocery	5

customer_churn 40

➢ a) Create a ‘customer_returns’ table in the ‘ecomm’ database and insert the following data:

```
6023 •  desc customer_returns;
```

Result Grid | Filter Rows: | Export: | Wrap Cell C

Field	Type	Null	Key	Default	Extra
ReturnID	int	NO	PRI	NULL	
CustomerID	int	YES	MUL	NULL	
ReturnDate	date	YES		NULL	
RefundAmount	int	YES		NULL	

Result 42

```
6035 •  select * from customer_returns;
```

Result Grid | Filter Rows: | Edit:

ReturnID	CustomerID	ReturnDate	RefundAmount
1001	50022	2023-01-01	2130
1002	50316	2023-01-23	2000
1003	51099	2023-02-14	2290
1004	52321	2023-03-08	2510
1005	52928	2023-03-20	3000
1006	53749	2023-04-17	1740
1007	54206	2023-04-21	3250
1008	54838	2023-04-30	1990
NULL	NULL	NULL	NULL

customer_returns 44

b) Display the return details along with the customer details of those who have churned and have made complaints.

```
6041    join customer_churn cc on cr.customerid = cc.customerid
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ReturnID	CustomerID	ReturnDate	RefundAmount	CustomerID	Tenure	PreferredLoginDevice	CityTier	WarehouseToHome	PreferredPaymentMode	Gender	HoursSpentOnApp	NumberOfDe
1002	50316	2023-01-23	2000	50316	0	Computer	2	29	UPI	Female	2	3
1004	52321	2023-03-08	2510	52321	18	Mobile Phone	3	19	E wallet	Male	2	1
1006	53749	2023-04-17	1740	53749	1	Mobile Phone	3	31	Credit Card	Female	3	5

Result Grid | Form Editor | Field Types

Result 45 | Read Only